Exercise 9.1

Given a dataset $\mathcal{D} := \{(a_i, b_i) \in \mathbb{R} \times \mathbb{R} \ : \ i \in \{1, \dots, N\}\}$ generated by

```
import numpy as np
import random
random.seed(10)

# generate data set
N = 100
inputs = np.random.uniform(0, 1, size=N)
outputs = 10 * inputs + np.random.normal(size=N)

# visualize data set
import matplotlib.pyplot as plt
plt.plot(inputs, outputs, '*')
```

We consider the optimization problem

$$\min_{x \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^{N} f_i(x)$$

with $f_i(x) = \left(\frac{1}{2}(a_i x - b_i)^2\right)$ for $i \in \{1, \dots, N\}$. Implement

(i) the stochastic gradient descent method with $N \cdot$ `max_epochs` iterations, and $t_k = \frac{1}{\sqrt{k+1}}$,

(ii) the stochastic gradient descent method with `max_epochs` epochs, N iterations per epoch, and $t_k = \frac{1}{\sqrt{k+1}}$,

(iii) the stochastic gradient descent method with `max_epochs` epochs, N iterations per epoch, and $t_k = \frac{1}{\sqrt{N+1}}$,

Choose `max_epochs` $= 10$ and `max_epochs` $= 1000$. Plot the iteration count vs. the function value of the ergodic mean.

Exercise 9.2

Consider the convex quadratic function

$$f(x) = \frac{1}{2N} \sum_{i=1}^{N} \|x - w^i\|_2^2,$$

with $x, w^i \in \mathbb{R}^n$, $i \in \{1, \ldots, N\}$, $\sum_{i=1}^N w^i = 0$, $\|w^i\|_2 = 1$ for $i \in \{1, \ldots, N\}$. Consider the stochastic gradient iteration defined by

$$x^{k+1} = x^k - \alpha_k(x^k - w_{i_k}),$$

where i_k is selected i.i.d. uniformly from $\{1, \ldots, N\}$, for some steplength $\alpha_k > 0$, where x^0 is any initial point.

1. Show that the minimizer of f is $x^* = 0$.
2. Express the conditional expectation $\mathbb{E}_{i_k}[\|x^{k+1}\|_2^2 | x^k]$ in terms of $\|x^k\|_2^2$ and α_k.
3. By applying the bound in (b) recursively and using the notation $A_K := \mathbb{E}[\|x^K\|_2^2]$, find a bound for A_K for any $K \in \{1, 2, \ldots\}$ in terms of $A_0 = \|x^0\|_2^2$ and $\alpha_0, \alpha_1, \ldots, \alpha_{K-1}$, where $\mathbb{E}$ denotes the expectation with respect to all random variables $i_0, i_1, i_2, \ldots$.
4. If the step length is chosen
   (i) constantly, i.e. $\alpha_k = \alpha > 0$
   (ii) such that $\alpha_k = \frac{1}{k+2}$

   for all $k \in \mathbb{N}$, does it hold that $\mathbb{E}[\|x^K\|_2^2] \to 0$ for $K \to \infty$?